

Assignment 1 memo — finance-industry job postings

Group topic: job market, split by industry • **Subtopic:** finance industry **Corpus:** 200 records from 175 postings, 27 employers, 3 public APIs; 74% carry a table **Extraction task:** 6 fields — `role_family` (12-value enum), `seniority` (6), `work_arrangement` (4), `certifications` (list), `required_skills` (list), `min_years_experience` (optional number) **Evaluation sample:** 28 records — 25 drawn at random (seed 0), plus 3 added to cover a cluster the draw missed

1. Human vs. SLM (Phi-4-mini-instruct)

Labelled against a written guideline before Phi was run. Mean latency 17.0 s per record on an M-series GPU.

Field	Accuracy	What goes wrong
<code>certifications</code>	86%	Misleading. 24 of the 28 records name no credential and Phi correctly returns <code>[]</code> . On the 4 that do name one — a law licence, CAMS/CFE, a NY bar admission, a registered architect — it returned an empty list every time. It never proposes a credential at all, so F1 on this field is 0.00.
<code>role_family</code>	75%	4 of the 7 errors collapse into a single class, <code>operations_and_client_service</code> . It is Phi's default bucket: an analytics engineer, an administrative assistant, a workplace-services associate and a systems administrator all land there.
<code>work_arrangement</code>	71%	Misses explicit statements. Stripe's "you will be required to work from the Chicago office three days per week" was read as <code>unspecified</code> .
<code>min_years_experience</code>	68%	—
<code>seniority</code>	43%	The worst field, 16 errors. 7 are upward drift , 5 of them mid answered as <code>senior</code> : Phi anchors on the stated years of experience, the guideline anchors on the title.
<code>required_skills</code>	0%	Scored by exact set equality, which nothing survives. Phi copies whole requirement sentences verbatim; one record returned 15 items, most of them full sentences. Precision 0.07, recall 0.14, F1 0.093.

`all_fields_correct_rate` is **0.0** for all 28 records, entirely because of `required_skills`. It is not a usable headline number for this task; the per-field view is.

Structured-output validity: 16 of 28 responses (57%) violated the schema. Two distinct failures:

- **14 records** wrote `"min_years_experience": null` where the field is optional and should simply be absent from the object.
- **2 records** answered `seniority: "unspecified"` — a value that does not exist in that enum. Phi borrowed it from `work_arrangement`, the neighbouring field that does have it. Enum bleed between adjacent fields.

What Phi handles well: closed enums where the document states the answer outright, and the negative case — returning [] when a posting names no credential. **What it handles badly:** judgements the document does not spell out (seniority from a title), abstraction rather than copying (a skill phrase from a requirement sentence), and omitting an optional field.

2. Recovery strategy

Target: the null-instead-of-absent violation — 14 of the 16 schema violations, unambiguous, and with an obvious prompt remedy.

Change (v2_recovery; all four prompt versions are stored in full in taxonomy.json): three instructions added to the strict prompt — omit the key entirely when no number is stated and never write null, 0 or "N/A"; copy enum values character for character and never borrow a value from another field; return one JSON object and nothing else.

	v1	v2_generated	v2_recovery	v3_skills
role_family	75%	79%	79%	82%
seniority	43%	39%	61%	54%
work_arrangement	71%	71%	71%	71%
certifications	86%	86%	86%	86%
min_years_experience	68%	64%	64%	68%
required_skills (exact set)	0%	11%	11%	4%
required_skills (F1)	0.093	0.083	0.065	0.228
Schema violations	16/28	17/28	14/28	17/28
— of which null	14	15	14	17
— of which out-of-enum	2	2	0	0

v2_generated is the control: the strict rewrite extraction.py builds automatically from the taxonomy, which tells the model to follow the rules exactly but names no specific failure. v2_recovery is the same base prompt plus the three targeted sentences. Nothing else differs between the two columns.

The targeted instruction did not work. The null count is 14 before and 14 after. Told three different ways to leave a key out, Phi left it out zero additional times; under v3_skills the same instruction produced 17. A 3.8B model appears unable to represent "do not emit this field" while generating an object whose schema lists it — a capability limit rather than a prompting problem, which §6 confirms against a 120B model that omits the key unprompted. The practical fix is to accept null in the schema, not to keep rewriting the prompt.

The secondary instruction did work, and targeting is what made the difference. Naming the enum values and forbidding cross-field borrowing took out-of-enum answers from 2 to 0 and lifted seniority from 43% to 61%. The generic strict prompt, given the same records, left both enum violations in place and pushed seniority slightly down, to 39%. Telling a small model to "follow the rules exactly" bought nothing; telling it that seniority has no value called unspecified bought 18 points on the worst field.

The asymmetry inside v2_recovery is the finding: **constraining what the model may output works, and asking it to output nothing does not** — and the comparison with v2_generated shows the constraint has to name the actual failure.

v3_skills attacked the other failure — each item a noun phrase of at most four words, verbs and qualifiers stripped. The output changed exactly as instructed: Anchorage's 15 copied sentences became ["digital asset trading agreements", "prime brokerage agreements", "collateral agreements", ...]. Exact-set accuracy nonetheless **fell** from 11% to 4%, while F1 rose from 0.065 to **0.228** and recall from 0.07 to 0.32.

The two metrics move in opposite directions because exact set equality cannot distinguish a wrong answer from a differently-worded right one. On the SoFi record v3 returned financial crime compliance, AML, sanctions, risk management, regulatory developments... against a human label of aml, sanctions, financial crime compliance, cryptocurrency and blockchain products, bsa/aml... — three exact overlaps, no disagreement of substance, scored as total failure. **For open-ended list fields the metric, not the model, was the binding constraint.** v3 also raised null violations from 14 to 17 and emitted a duplicate list item, neither of which any instruction addressed.

3. Cluster reading

k-means over TF-IDF, k = 6, **silhouette 0.061** — weakly separated, which is itself a finding: job postings share heavy boilerplate (benefits, EEO statements, company blurbs) that dominates the vector space.

Cluster	Name	Corpus n	Eval n	Field accuracy
2	Trading & quantitative research	49	3	67%
3	NYC legal & claims	14	2	67%
0	NYC budget & fiscal analysis	14	3	56%
1	US fintech product & risk	21	3	56%
5	Private-sector mixed	78	14	56%
4	NYC tax & administration	24	3	50%

Clusters 0, 3 and 4 are the public-sector half, split cleanly by function (budget, legal and claims, tax administration). Cluster 2 is the trading firms, cluster 1 the US fintechs; cluster 5 is a 78-record residue the clustering never resolved.

Performance tracks document structure, not subject matter. Cluster 2 scores best because trading postings are fully structured — title, years of experience, and an explicit requirements section. Cluster 4 scores worst because most of its records are NYC description-only records: the NYC dataset keeps qualifications in a separate column, which became a separate record, so these documents state no requirements, no work arrangement and no years. Phi fills the gap by guessing, and guesses in a fixed direction — senior for seniority, operations_and_client_service for role family.

Cluster 0 is the same effect running the other way. Its OMB budget postings describe "the City government's chief financial agency" and use the word "budget" a dozen times, and Phi labelled role_family correctly on **all three** — its best result for that field on any cluster. It then lost most of those

records on seniority, because "Deputy Assistant Director" and a three-grade posting ("Assistant Analyst / Analyst / Senior Analyst") are exactly the judgements it cannot make.

Two caveats. The original random draw left cluster 0 with **no** evaluation records, so three were added deliberately to cover it; the sample is therefore not a pure random draw, and `human_labels.jsonl` flags those three. And with 2–14 records per cluster the ordering is indicative only.

4. Corpus limitations

The corpus is a one-day snapshot of US — largely New York — finance hiring, so it cannot answer anything about trends, seasonality, or non-US labour markets. Employers recruiting through Workday or Taleo are absent entirely, which removes every large bank and insurance carrier: there is no Goldman Sachs, JPMorgan or BlackRock posting here, and `investment_banking` and actuarial roles are effectively unrepresented. Salary appears only where a pay-transparency law or the NYC dataset supplies it, so any compensation analysis is biased toward four US states. Postings describe what employers advertise, not who was hired or at what pay, so questions about outcomes are out of reach. Finally, the per-employer cap of 8 makes record counts a diversity choice, not a measure of hiring volume.

5. Two notes on method

Sensitive information. A sweep of all 200 records found 21 email addresses, all organisational mailboxes, no phone numbers and no personal identifiers, so there was nothing to mask. Phi was run over the flagged records anyway, as §2.1 asks. It proposed 81 spans, **none of them personally identifiable**, labelling job titles `<PHI>`, employer names `<PII>` and whole responsibility bullets `<CONF>` wherever a sentence contained the word "confidential"; ten of its 76 calls returned no parsable JSON. The output was discarded and the corpus ships unmasked. This is an independent replication of §1's finding on the same model and the same documents: Phi is usable for copying and for closed classification, and not for judgement.

6. Bonus: SLM vs. LLM

The same 28 records, the same `v1_original` prompt and the same schema, sent to `openai/gpt-oss-120b` on Groq — roughly 31 times Phi-4-mini's parameter count. (The starter's default, `llama-3.3-70b-versatile`, has been retired from Groq's catalogue and now returns a 404; `llm_utils.py` records the substitution.)

	Phi-4-mini (3.8B)	Phi, v2_recovery	gpt-oss-120b (v1 prompt)
role_family	75%	79%	89%
seniority	43%	61%	68%
work_arrangement	71%	71%	86%
certifications	86%	86%	86%
min_years_experience	68%	64%	68%
required_skills (exact set)	0%	11%	4%
required_skills (F1)	0.093	0.065	0.279
Schema violations	16/28	14/28	0/28
— of which null	14	14	0
Mean latency	17.0 s	—	9.6 s

The larger model settles the question §2 could not. Given the *baseline* prompt — no instruction about omission at all — gpt-oss-120b left `min_years_experience` out of the object 15 times and produced **zero** schema violations. Phi, told three separate ways to omit that key, wrote `null` 14 times out of 14. Emitting a legal optional-field JSON object is therefore within reach of a 120B model and outside the reach of a 3.8B one at any prompt we tried: a capability limit, as §2 inferred, now with a control rather than an inference.

Judgement errors are shared, and only partly a scale effect. The LLM's `seniority` mistakes have the same shape as Phi's — four mid answered senior, two lead answered senior, one executive answered senior. It is 25 points better than Phi and still drifts upward in exactly the same direction, which suggests part of that disagreement is the guideline's title-over-years rule being genuinely contestable rather than the model being small. That is a caution about the reference labels, not only about the models.

certifications shows the metric failing, not the model. Both systems score 86%, but for opposite reasons. Phi returns `[]` on all four records that name a credential — it never proposes one. The LLM finds two of the four and words them differently: NY Bar Admission against the label New York State law licence, and New York State Architect Registration against New York State Registration as an Architect. Both are correct and both are scored wrong. The same effect drives the `required_skills` gap: the LLM triples Phi's F1 (0.279 vs 0.093) while its exact-set accuracy stays near zero.

What this means for the corpus. Extraction over these documents is viable with a hosted model — 89% on a 12-way role classification, 0% schema violations, 9.6 s per record — and not viable with Phi-4-mini for anything but the closed, explicitly-stated fields. The fields that remain hard for both, `seniority` and `required_skills`, are hard for reasons this corpus cannot fix: postings state neither a seniority level nor a canonical skill vocabulary, so any evaluation of them is measuring the annotation guideline as much as the model.